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Solving the Travelling Salesman Problem using Agentic AI

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Outline:

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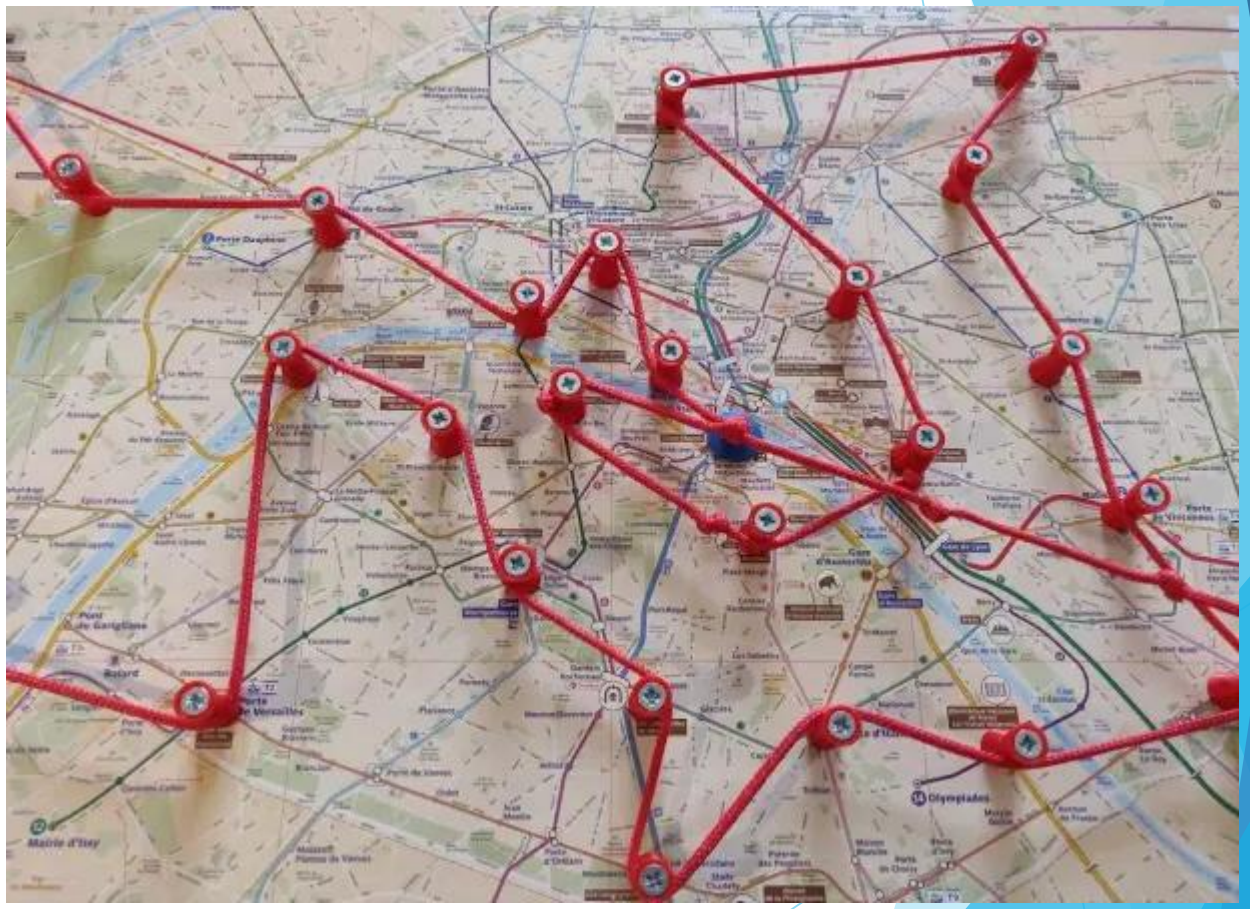
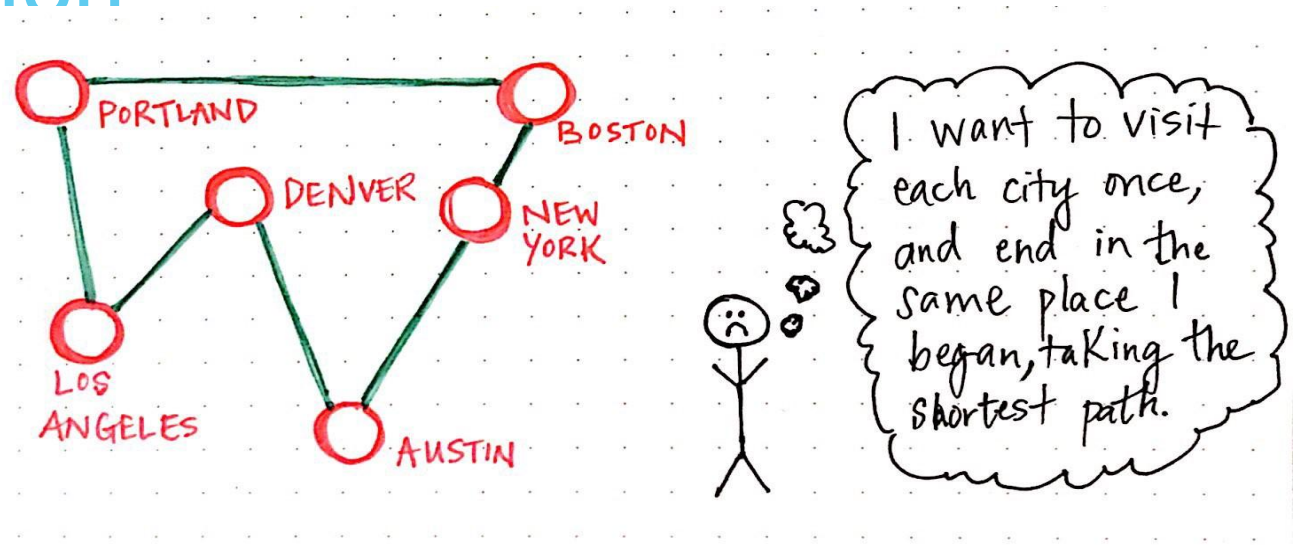


Introduction

- The Travelling Salesman Problem (TSP) is a classical combinatorial optimization problem that aims to find the shortest possible route that visits each city exactly once and returns to the starting city.

Introduction



- TSP is classified as an NP-hard problem because the number of possible routes grows factorially with the number of cities (n!). As the problem size increases, traditional exact algorithms become computationally infeasible.

Literature Review

- TRADITIONAL METHODS:

  - Exact algorithms: Brute Force, Dynamic Programming, Branch & Bound
  - Guarantee optimal solutions
  - Computationally infeasible for large-scale TSP

Literature Review

- SOFT COMPUTING:
- Metaheuristics:
  - Genetic Algorithm (GA)
  - Ant Colony Optimization (ACO)
  - Particle Swarm Optimization (PSO)
- Provide near-optimal solutions efficiently
- Suitable for large problem sizes

Literature Review

- AGENTIC AI:
- Agentic AI coordinates autonomous agents
- Enables task decomposition and solver orchestration
- Frameworks like **jMetal** support scalable optimization *[1]*
- Effective for hybrid TSP solving *[2]*

Problem Statement

- Finding optimal routes for large-scale TSP is computationally difficult
- Traditional methods lack scalability and adaptability

Objectives

- To apply **Agentic AI** for solving large-scale Travelling Salesman Problems.
- To enable **autonomous planning and optimization** using multiple intelligent agents.
- To use **soft computing algorithms** for efficient route finding.
- To achieve **faster and adaptive** solutions compared to traditional methods.

Proposed Solution: Overview

- Agentic AI-based hybrid optimization framework[2] [3]
- Uses multiple Optimization Methods
- Compares individual and hybrid approaches

Proposed Solution: Decomposition

- Greedy → Initial Solution
- GA → Improves Solution
- ACO → Refines Path
- Combined into hybrid model

Proposed Solution : Local Optimization

- Optimization agents can apply:
  - GA
  - ACO
  - Greedy[1]
- Near-optimal routes are targeted for each cluster

Proposed Solution: Coordination & Learning

- Methods Tested:
  - Greedy, GA, ACO
  - Greedy + GA, GA + ACO , Full Hybrid
- Compared Using
  - Tour Length
  - Execution Time
- Checked the Best performing method

RESULTS:

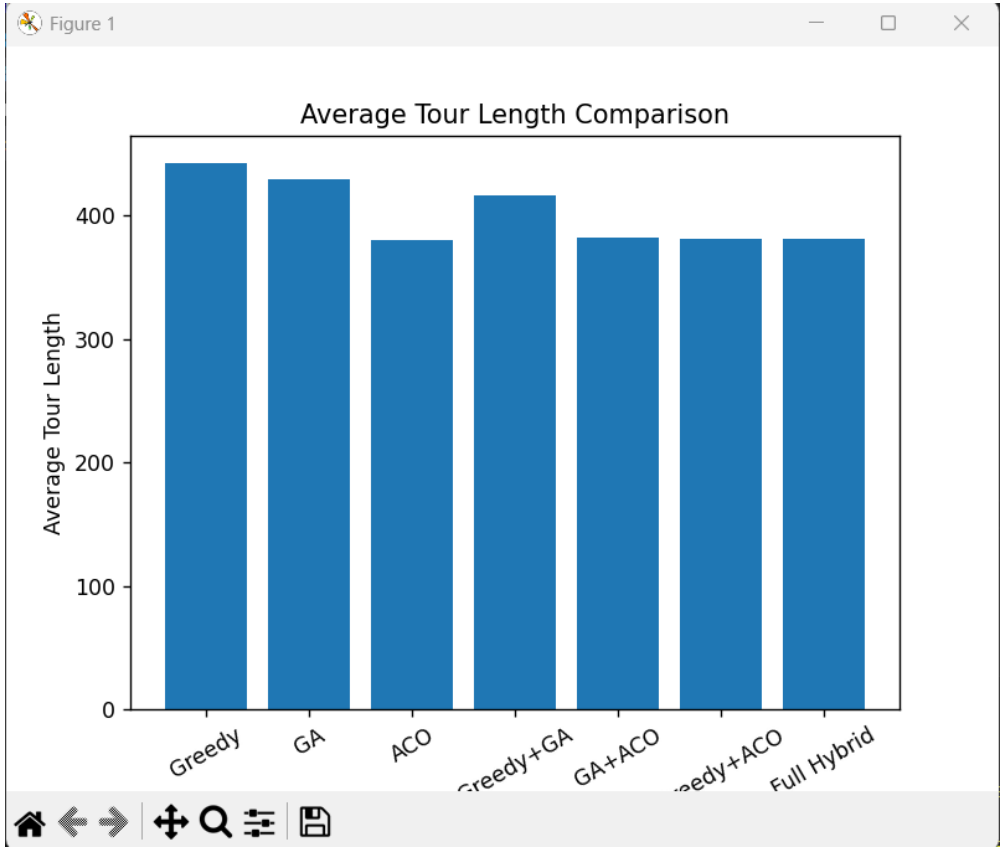


Figure 1
Average Tour Length Comparison
Average Tour Length
0
100
200
300
400
Greedy
GA
ACO
Greedy+GA
GA+ACO
Full Hybrid

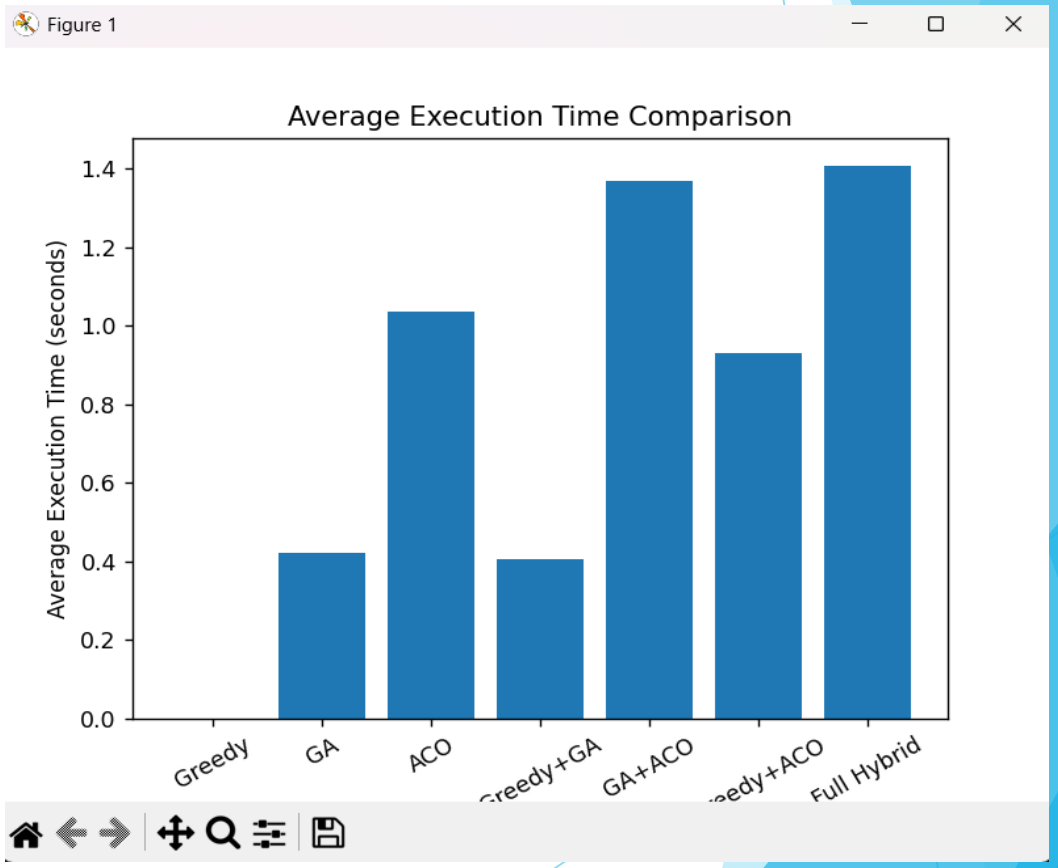


Figure 1
Average Execution Time Comparison
Average Execution Time (seconds)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Greedy
GA
ACO
Greedy+GA
GA+ACO
Full Hybrid

RESULTS:

- A Python-based simulation was developed for the Travelling Salesman Problem with **N cities**.
- Greedy , Genetic Algorithm (GA) , Ant Colony Optimization (ACO) approaches were implemented and experimentally evaluated.
- Greedy → Fastest Execution
- GA & ACO → Better Tour Lengths
- An **Agentic AI-based hybrid approach** was designed by combining Greedy initialization with GA refinement and ACO's optimization to balance **solution quality and execution time**.

OUTCOME:

- Designed a **agentic hybrid approach** for solving the classical TSP using Agentic AI as an orchestrator.
- Demonstrated that no single algorithm performs best in all scenarios, motivating a **dynamic hybrid solution**.
- Developed a working simulation framework in Python for experimentation and comparison.
- The system will achieve **improved solution quality and adaptive performance** compared to standalone algorithms.

References:

[1] C. A. C. Coello, J. J. Durillo, and A. J. Nebro, “jMetal: A Java framework for multi-objective optimization,” *Advances in Engineering Software*, vol. 39, no. 10, pp. 937–951, 2008.

[2] M. Wooldridge, *An Introduction to Multi-Agent Systems*, 2nd ed. Chichester, U.K.: Wiley, 2009.

[3] OpenAI, “Agentic AI: Autonomous and tool-using systems,” Technical Overview, 2024.

Thank You